

Library Management System

Software Analysis and Design (SAD) Requirements Report

1. Executive Summary

This report consolidates Requirements Engineering concepts applied to the Library Management System (LMS). It serves as a comprehensive technical guide for developers, stakeholders, and project reviewers, detailing the transition from manual processes to an automated digital ecosystem.

2. Problem Definition & Scope

The Problem: Inefficiencies in manual operations, specifically tracking borrowed books, managing returns, handling fines, and maintaining inventory accuracy.

System Boundaries

In Scope	Out of Scope
• Catalog Search & Member Registration • Borrowing, Returns & Reservations • Fine Management & Reporting • Librarian & Admin Operations	• Payment Gateway Integration • Multi-branch Synchronization • RFID Automation • Advanced Recommendation Engines

3. Stakeholder Analysis

Identifying the key actors and their specific requirements within the system.

- **System Administrator:** Configure policies, define fine rates, manage staff roles, and maintain system-wide controls.
- **Librarian:** Issue/return books, manage inventory, register members, and handle overdue cases.
- **Member (Student):** Search the catalog, borrow/reserve books, and manage account obligations.

4. Functional Requirements (FR)

The core features that define the system's behavior.

ID	Requirement Description
FR1	Enable catalog search by title, author, and genre.
FR2	Validate member eligibility (e.g., existing fines, limit reached) before borrowing.
FR3	Generate unique transaction records for every borrowing event.
FR4	Immediate update of book availability status upon issue or return.
FR5	Automated calculation of overdue days and support for fine processing.
FR6	Support for placing holds/reservations on unavailable books.
FR7	Generate management reports for borrowed books, overdue items, and inventory.

5. Non-Functional Requirements (NFR)

The quality attributes that ensure the system is robust and professional.

Category	Requirement
Performance	Search and transaction responses must complete within acceptable industry thresholds.
Availability	System uptime must be maintained during all operational hours with minimal downtime.
Security	Implementation of Role-Based Access Control (RBAC) to restrict unauthorized actions.
Usability	Intuitive UI allowing librarians and members to perform core tasks with minimal training.

Reliability	Ensuring data integrity and consistency across transactions and inventory records.
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6. System Modeling & UML

Use Case Overview



Core Workflows

- **Borrow:** Search -> Eligibility Check -> Validate -> Record Transaction -> Update Inventory.
- **Return:** Return Entry -> Due Date Check -> Overdue Logic -> Update Inventory -> Close Transaction.

Data Model (ER Model)

Entities identified for the database: **Member, Book, Transaction, Reservation, Fine, and Librarian.**

7. Implementation Mapping

The proposed technology stack for the prototype architecture.

Frontend: React.js	Backend: Node.js / Express	Database: MySQL
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8. Risk & Validation

- **Validation Criteria:** Validity, Consistency, Completeness, Realism, and Verifiability.
- **Validation Methods:** Walkthrough reviews, prototype checks, and scenario test cases.
- **Risks:** Ambiguous requirements, incomplete stakeholder input, and scope creep.

